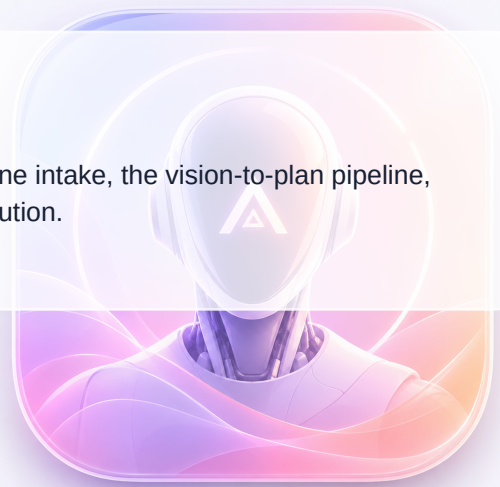


See the board. Plan the action. Move with intent.

A visual, practical guide to A3Terra: the glass-surface desktop environment for understanding a real work area, preparing robot tasks, rehearsing them, and optionally sending the verified command sequence to hardware.

IN THIS GUIDE

New in A3Terra compared with A2.6 Soil, fast setup, robust scene intake, the vision-to-plan pipeline, simulation, voice, configurable controls, and guarded USB execution.



What A3Terra is

A3Terra turns a photo or camera frame into an operating board: physical objects are identified, localized against a 20 x 11 grid, considered against a natural-language task, and then represented as a visible command sequence.

01

Bring in a scene

Import a photo or start a USB camera feed.

02

Measure it

Vision maps objects to usable board cells.

03

Describe work

Type or dictate the outcome you want.

04

Review first

Inspect and simulate before any real execution.

● A glass-first operating surface

A3Terra uses translucent panels, subtle borders, gentle shadows, and high contrast text to keep the image, grid, conversation, and execution state legible together.

● Clear separation of concerns

Scene analysis, task generation, playback, hardware connection, and operator settings are separate controls. A connected cable alone never authorizes a command to leave the app.

Core operating principle

Treat the board image as the current source of truth. If the view changes, re-import or restart the camera feed so the next analysis is based on the actual scene rather than stale geometry.

What is new in A3Terra

A3Terra is not just a new skin for the A2.6 Soil simulator. The underlying workflow becomes more measurement-aware, resilient to real photos, and explicit about execution safety.

● Measured image canvas

Images are letterboxed to a true square canvas and annotated with a 0-1000 ruler before vision analysis. This corrects aspect-ratio bias when converting visual locations into board coordinates.

● More reliable object coverage

Polygon scoring samples cell coverage, keeps only enough of an object within the photo, limits runaway claims, and separately rejects background-sized regions. This makes board overlays more trustworthy in ordinary scenes.

● Opt-in pixel snapping

Segmentation-based snapping is available for clean, high-contrast scenes, but is off by default. The model outline remains the safer baseline around soft shadows, reflective surfaces, and low contrast objects.

● Modernized operating stack

The supplied code names GPT-5.4 for vision, GPT-5.6-terra for planning, GPT-5.4-mini for dexterity, GPT-5.4-nano for voice tidy, and gpt-4o-transcribe for speech.

Start with a scene you can trust

A

Choose source

Import JPG, PNG, BMP, TIFF, or WEBP; or select a USB camera.

B

Confirm frame

Keep the full work area visible and avoid occlusion.

C

Automatic pass

Vision begins after a readable import or the first usable live frame.

D

Inspect overlay

Check the object outline and cells before requesting a task.

● Image intake is defensive

A3Terra normalizes unusual paths, grayscale images, RGB/RGBA files, and non-8-bit imagery. Transparent input is composited on white rather than silently losing alpha. These details help imported scenes behave predictably.

● The 20 x 11 operating grid

Board labels run A-T and 1-11. The ruler canvas is used during vision measurement, then coordinates are mapped back to the original image and drawn where the operator can verify them.

Good capture habits

Use even lighting, avoid hard glare, leave a margin around the working surface, and choose a viewpoint that makes the object boundary and intended destination visible. For complex scenes, prefer a clearer replacement image over trying to force a weak interpretation.

From request to board-aware plan

Write the outcome you want in the sidebar, such as "pick up the knife" or "move the bowl to an empty area." The app keeps the task grounded in the current image rather than treating it as an abstract text prompt.

1

Vision

Find discrete objects and outline their physical extent.

2

Parts

Inspect actionable features such as a handle, rim, or control.

3

Dexterity

Test whether the requested interaction is feasible.

4

Planner

Write a coordinate-aware command sequence.

● Ask plainly, add the goal

Include the object and destination when the destination matters: "Place the cup at the clear right side." The system can choose a known empty location only where that is a reasonable default.

● Missing is better than invented

If a requested item is not visible, A3Terra is designed to surface that limitation rather than fabricate a location. Use a clearer scene, reframe the camera, or name the visible object more specifically.

REVIEW MOMENT

A generated plan is a proposal. Make simulation your default rehearsal before USB execution.

Read the scene, then refine it

● Second-pass verification

Keep outline verification on when tasks depend on precise placement. The second look is particularly valuable where an object edge is thin, a shadow is attached to its base, or nearby objects overlap.

● Snap outlines to pixels

Use snapping only on a simple contrasting surface where objects are well separated. It is intentionally per-scene and experimental: a confident but wrong contour is less useful than a slightly approximate model outline.

● Background rejection

Large regions that cover much of the frame or have insufficient foreground evidence are treated as scene background. Named large subjects - a bed, car, sofa, or appliance - are treated more carefully so they are not removed merely for being large.

● Board coverage tuning

The app uses a touch threshold, relative fallback, padding retention, maximum claimed cells, and multi-sample polygon coverage. These settings influence how a visual outline becomes a practical cell footprint.

Before trusting a plan, ask: does the outline contain the intended thing? Does the highlighted footprint match its real extent? Is the contact point on a safe, meaningful part of the object? If not, fix the scene first.

Use the fastest input for the moment

● Voice dictation with safeguards

A3Terra records from the default microphone, reports a blocked or silent input clearly, and rejects audio too quiet to transcribe. The recorder is designed not to send a half-thought merely because the speaker pauses.

● Transcription and tidy pass

Speech is transcribed off the interface thread, with a fallback path if the primary transcription route fails. A cleanup pass can remove fillers before the text is placed in the task field.

● Standing instructions

Custom rules live in the editable AI_INSTRUCTIONS block. Write precise, conditional guidance - for example, a condition on a task involving soap - rather than a broad rule that could distort every future plan.

● Example scenes

Use the examples menu to load a prepared task and scene together. It is the fastest way to learn the visual cues, inspect a full plan, or retest a familiar task after changing an advanced setting.

Practical dictation pattern

Say the desired result first, then the object, then the destination or constraint: "Move the blue cup to the clear left area" or "Pick up the knife by the handle." Edit the transcript before sending when a word was misheard.

Rehearse the outcome before hardware

01

Simulate

Run the exact parsed commands over the board overlay.

02

Observe

Watch the target, motion, grip, placement, and waits.

03

Adjust

Stop, revise the scene or request, then regenerate.

04

Re-run

Repeat the last sequence when the visual result deserves another look.

● Playback controls

Set a live speed multiplier. Long waits can be reported accurately while a simulation cap keeps review practical. Stop interrupts the current run; label visibility can be toggled when you need a cleaner or more explicit board.

● What to check visually

Verify the intended outline, path and contact cells; confirm the grip makes sense; look for a clear destination; and ensure any waits, releases, or subsequent movements still match the physical goal.

COLOUR LEGEND

Use the colour legend whenever an overlay state is unclear. Turning labels off can declutter a board, but the legend remains the quickest way to

Connect intentionally. Arm intentionally.

Hardware sending is deliberately separated from planning and simulation. A port must be connected, and the explicit "Send generated commands over USB" switch must be enabled, before the generated sequence is written to the device.

1**Find port**

Refresh the list of serial devices and select the intended port.

2**Set baud**

Choose the matching baud rate; 115200 is the source default.

3**Connect**

Confirm the link without automatically arming command sending.

4**Arm**

Enable the switch only for a plan you have reviewed.

● One plan, one wire representation

The serial link uses the same command parser as the simulator. It writes only the bare command lines - not step numbers, comments, or missing notices - exactly as the canvas execution understands them.

● Keep calibration separate too

AprilTag calibration should be completed before relying on the board mapping for real-world execution. Connection proves the cable works; calibration aligns the camera, board, and physical destination.

If a port is connected but the arm switch is off, the app should report that it is connected and simulation-only. If the switch is on but no port is open, it should warn that nothing can be sent. Those two checks are intentional safety boundaries.

Settings, recovery, and a safe routine

● Useful controls to know

Simulation: speed, wait cap, verification, snapping, labels.
Detection: touch, padding, background, coverage. Network: timeout, retries, backoff. Each setting should be changed with a purpose and validated on a known scene.

● Model and UI defaults

The supplied code currently assigns GPT-5.4 to vision, GPT-5.6-terra to the planner, GPT-5.4-mini to dexterity, GPT-5.4-nano to dictation tidy, and gpt-4o-transcribe to speech. The light glass palette uses cyan, blue, violet, pink, green, amber, and red status accents.

● Fast recovery

If results look wrong, stop the run, inspect the overlay, replace or reframe the scene, then try again. If calls fail, check the key and network settings. If you lose track of adjustments, restore defaults and revalidate with an example.

● Keyboard shortcuts

Ctrl+I imports an image. Ctrl+Shift+C connects a USB camera. F11 enters or leaves fullscreen; Esc leaves fullscreen. The right Option key on macOS is used for dictation and send where configured.

THE OPERATING CHECKLIST

Clear scene -> sound outline -> clear task -> review plan -> simulate -> calibrate -> connect -> arm -> supervise.

Guide basis: supplied A3Terra source code and the provided A2.6 Soil code excerpt. Visual direction: source UI glassmorphism and supplied reference artwork.